

Pultrusion of structural components made of low-cost PAN carbon fiber composites



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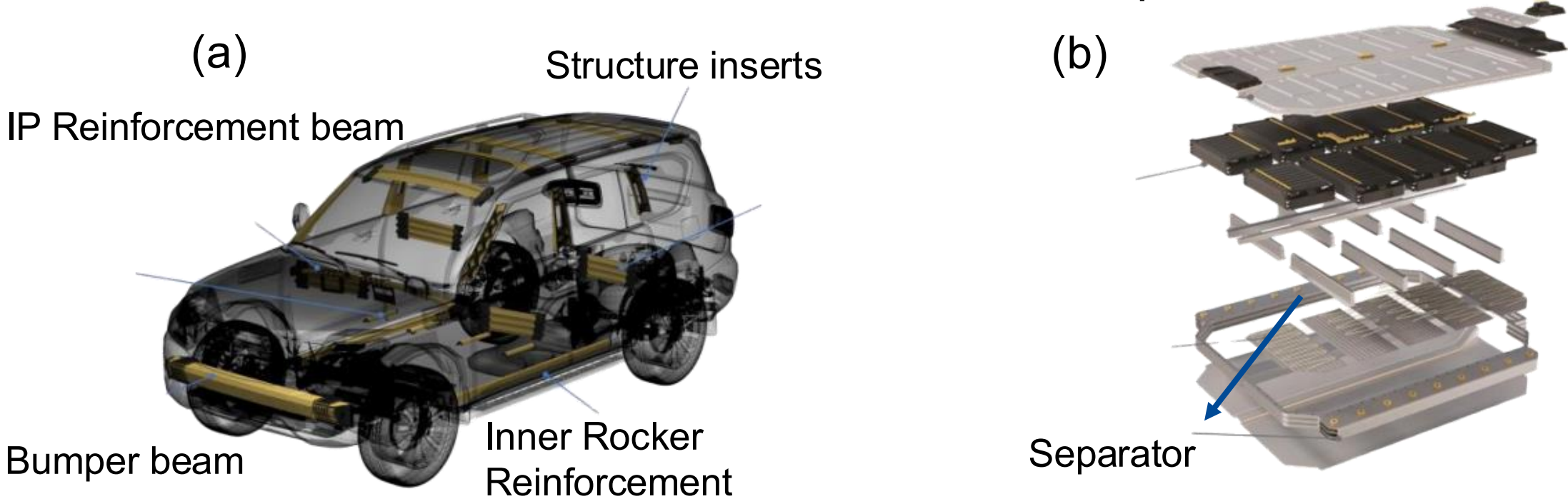
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BACKGROUND & OBJECTIVE

>Traditional Battery Module Separator Manufacturing

- **Weight:** Classic metal materials add significant weight to vehicles.
- **Emissions:** High CO2 output during processing & manufacturing.
- **Cost:** Traditional metals and carbon fiber are expensive.



Examples of structural beams of (a) a Toyota model (courtesy of Toyota), and (b) electrical battery separators Source: ChemistryWorld

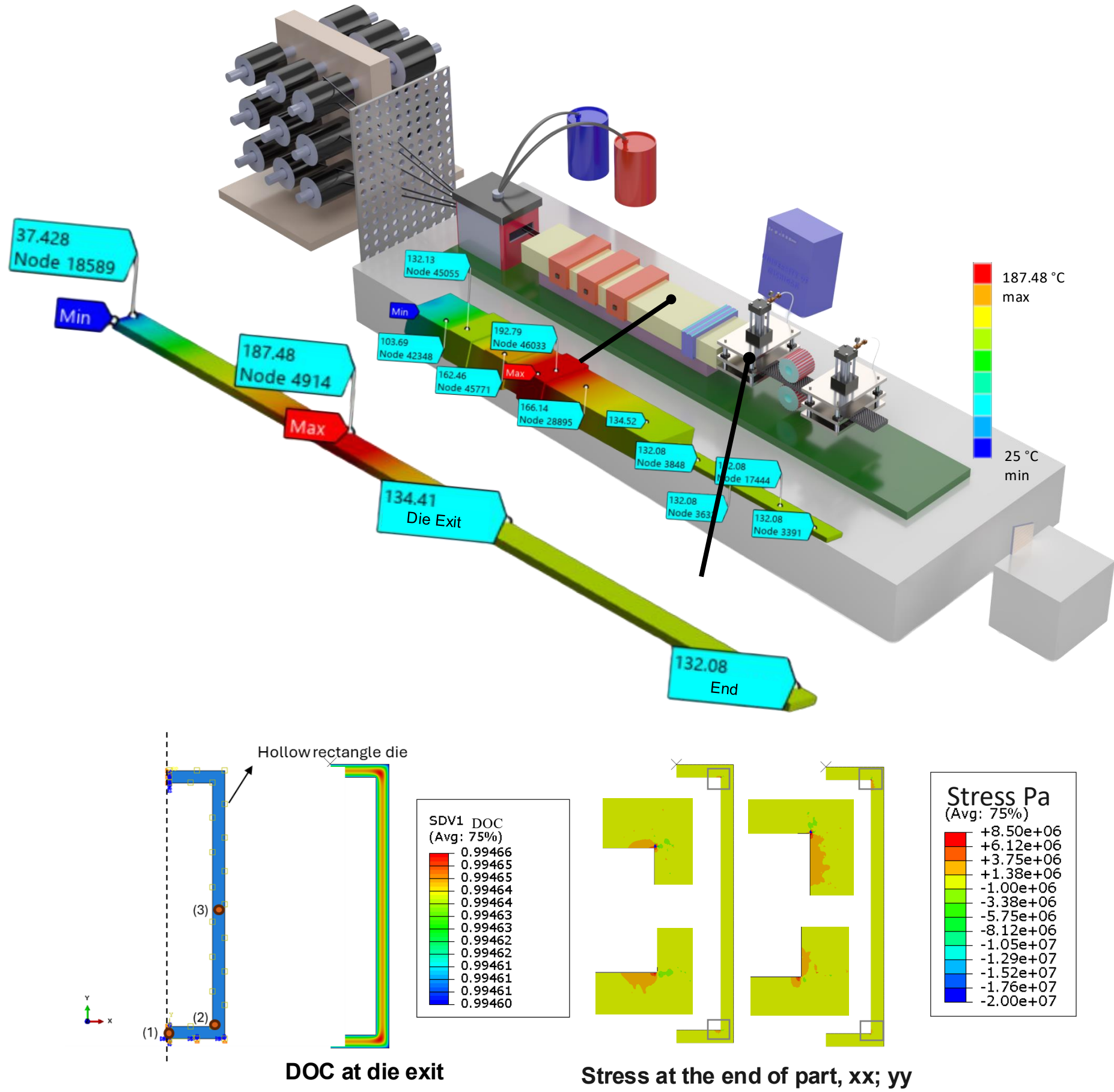
>Pultrusion Process for Carbon fiber composites:

A continuous, automated manufacturing method used to produce fiber-reinforced composite parts with constant cross-sections.

Objectives:

- 50% increase in specific strength compared to steel;
- 60% weight reduction based on steel;
- 25% reduction in greenhouse gas emission.

SIMULATION-GUIDED PERFORMANCE DIGITAL TWIN OF A PULTRUDER



Curing & Residual Stress at die exit

- Avg. degree of cure: **99.46%**
- Max residual stress: **~8.5 MPa**
- Avg. shrinkage: **1.23%**

Temperature is sum of two sigmoid functions:

$$T(x) = 25 + \frac{T_{peak1} - 25}{1 + e^{-(20(x-0.3))}} + 25 + \frac{T_{peak2} - 25}{1 + e^{-(20(x-0.7))}} + 273.15$$

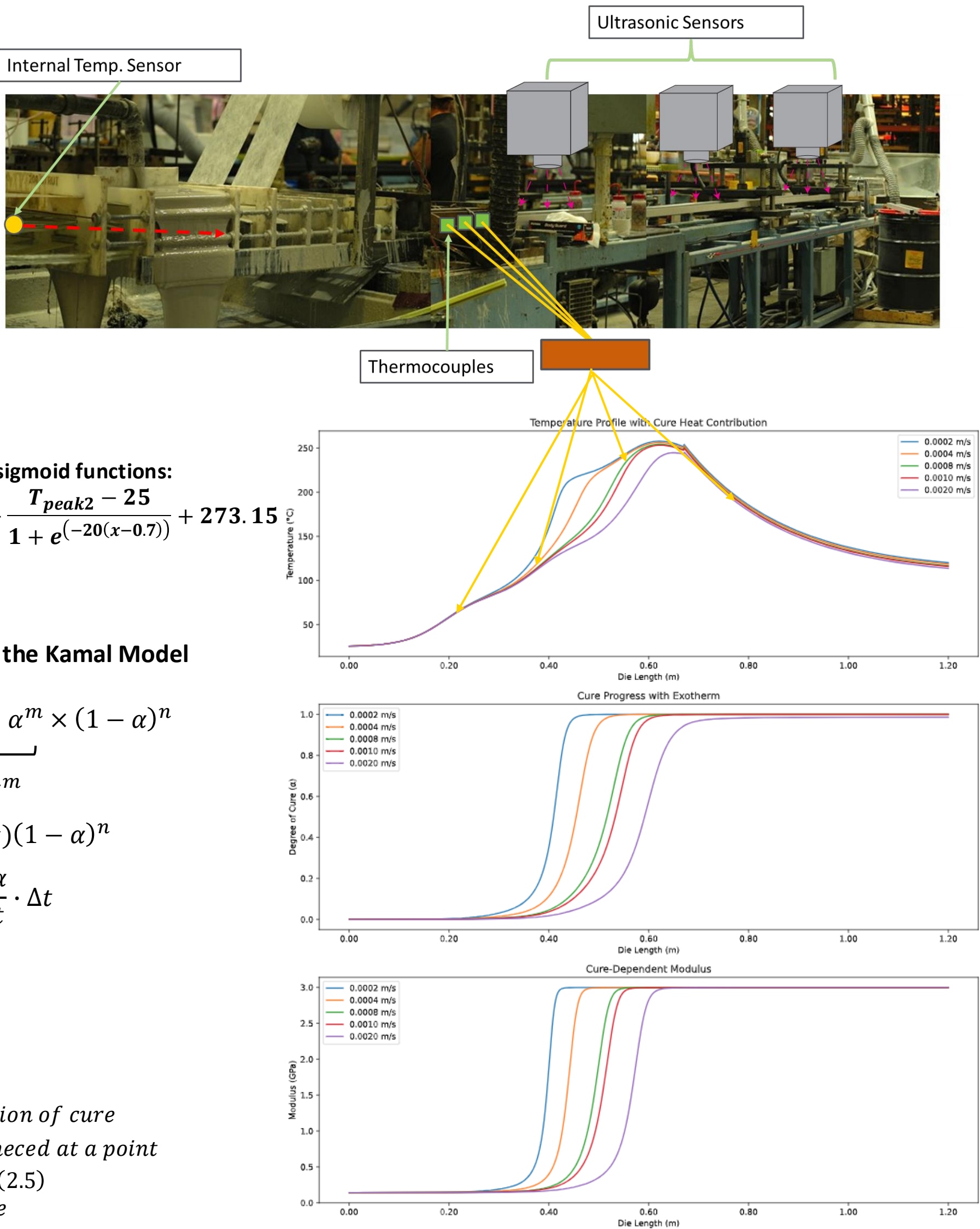
Cure rate computed from the Kamal Model

$$\frac{d\alpha}{dt} = A_1 e^{-\frac{E_1}{RT}} + A_2 e^{-\frac{E_2}{RT}} \cdot \alpha^m \times (1 - \alpha)^n$$
$$\frac{d\alpha}{dt} = (k_1(T) + k_2(T)\alpha^m)(1 - \alpha)^n$$
$$\alpha_{t+1} = \alpha_t + \frac{d\alpha}{dt} \cdot \Delta t$$

Modulus $E(\alpha)$

$$E(\alpha) = E_0 \cdot \alpha^n$$

$E(\alpha)$: Modulus as a function of cure
 α : Degree of cure experienced at a point
 n : Power-law exponent (2.5) fitted from literature



Real-time monitoring & extracting data history plot during the pultrusion

CONCLUSIONS

- NHTSA Side pole test initialized the loading condition for fiber distribution.
- Sub-model and pixel-based structural analysis were built to optimize local mechanical behavior.
- Digital twin pipeline: **Design** → **Simulation** → **Manufacturing** → **Performance Prediction** are setting up.
- AI acts as the core engine to predict structural performance and optimize material distribution.

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